

Why the supervision camera shares the anchor

Coordinate control and refinement loop

Before a loop closure

Local map and camera use keyframe k.
Their relative transform is fixed.



$$T_{WCf}^{-1} T_{Wck} = T_{CkCf}^{-1}$$

Pose correction

The pose graph changes T_{Wck} .
The map and camera move together.



shared anchor: relation preserved

Controlled 6% scale test

Fixed world camera: refinement reverses the correction.
Shared anchor: correction remains after about 500 steps.

Desk + room controls

REFINEMENT LOOP

sample RGB

compose pose

render map

L1 + SSIM

Adam update

Only Gaussian scale, rotation, opacity, and DC color receive gradients.

